Indira Gandhi National Open University

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BPAG-171: Disaster Management
Assignment (TMA)

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This assignment consists of Section A, B and C. Answer all the questions, marks are indicated on the right hand side.

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Section-'A'

Que 1. Define the term 'disaster' and Discuss its typology.

Ans → The term 'disaster' refers to a Sudden, calamitous event that causes significant disruption to the normal functioning of a community or society, resulting in widespread human material, economic or environmental losses. Disasters over-whelm the capacity of the affected community to cope using its own resources, often necessitating external assistance for relief and recovery. In essence, a disaster is not merely a natural or human event; it becomes a disaster when it significantly impacts human life, property infrastructure and livelihoods.

Disasters are typically classified into different types based on their origin, nature and efforts. understanding the typology of disasters is crucial for effective disaster management, preparedness and mitigation Strategies. Broadly disasters can be categorized into natural, human-induced and complex emergencies.

1. Natural Disasters:

Natural disasters are caused by natural phenomena, often beyond human control and have a direct impact on human life and the environment. They can further be Sub-categorized as:

- Geological Disasters: These originate from the Earth's internal processes. Examples include earthquakes, volcanic eruptions, tsunamis and landslides. Earthquakes, Such as the 2001 Bhuj earthquakes in India, cause structural damage, fatalities and long-term socio-economic consequence

- Hydrological Disasters: These are related to water and its movement. Floods, Cyclones, Storm, Surges and droughts fall under this category. Floods, for instance are frequent in the Ganga-Brahmaputra plains and lead to loss of lives crops and infrastructure.

- Climatological Disaster: These are caused by long lived atmospheric processes. Extreme temperatures, heatwaves, cold waves and wildfires are example. Heatwaves in India, such as those affecting Andhra pradesh and Telangana result in severe human health impacts.

- Biological Disasters: These involve the spread of diseases caused by microorganisms, viruses or other biological agents. Epidemics, pandemics and infestations like locust swarms are included here. The COVID-19 pandemic exemplifies a large-scale biological disaster.

2. Human-Induced Disasters:

These disasters are caused directly or indirectly by human actions, often due to negligence, poor planning or deliberate acts. They can include:

- Technological Disasters: These results from the failure of industrial systems, machinery or infrastructure. Examples include chemical spills, nuclear accidents (like the Bhopal Gas Tragedy 1984) industrial explosions.

- Sociological Disasters: These arise from human conflict social unrest or political instability, Terrorist attacks communal riots and wars fall under this category. They not only cause immediate loss of life but also long-term socio-economic disruption.

- Environmental Degradation: Deforestation, soil erosion and industrial pollution may not cause immediate disasters but can lead to gradual environmental collapse increasing vulnerability to other disasters like floods or landslides.

3. Complex Emergencies:

These occur when natural and human-induced disasters intersect, creating compounded crises. For example, an earthquake leading to a tsunami that subsequently triggers a humanitarian crisis falls under this category. Complex emergencies often involve mutiple sectors - health, governance and social infrastructure - making disaster response challenging. The typology of disasters is essential for disaster management, which includes mitigation, prepadredness, response and recovery. Classifying disasters allows policymakers emergency responders and communities to develop tailored strategies, allocate resources effectively and implement early warning systems. In India, a country highly prone to both natural and human-induced disasters understanding disaster typology is crucial for reducing vulnerability and enhancing resilience at local, regional and national levels.

In conclusion, a disaster is a significant event that disrupts societal functioning, causing severe loss and requiring external intervention. Its typology - natural, human-induced and complex - helps in systematic planning, risk assessment and efficient management to safeguard lives, property and the environment.

Que. 2. Discuss disaster management policy of Government of India.

Ans – The Govt. of India has developed a comprehensive disaster management policy to address the challanges posed by natural and man-made disasters, emphasizing a proactive integrated and multi dimensional approach. The policy aims to minimize loss of life, reduce damage to property, and enhance the nation's capacity to respond effectively to emergencies. Disaster management in India has evolved significantly over the decades, shifting from a reactive relief oriented system to a more holistic, preparedness-based framework.

The national policy on disaster management, first formulated in 2009, provides the overarching framework for disaster management in the Country. It emphasizes a shift from relief - centric approaches to proactive risk reduction, preparedness, mitigation and community participation. The policy recognizes the multidimensional nature of disasters, which includes natural hazards like floods, earthquakes cyclones and droughts as well as technological and human - induced disasters. Its primary objectives include creating a safe and disaster - resilient India, developing a culture of prevention, preparedness and mitigation and promoting the integration of disaster management into all level of plannings and development.

A key feature of India's disaster management policy is the institutional framework established to ensure coordinated efforts at national, state and district levels. The apex body, the national disaster management Authority (NDMA)

Chaired by the prime minister is responsible for laying down policies plan and guidelines for disaster management NDMA coordinates with various ministries, State authorities and international agencies to implement disaster management Strategies effectively. At the State level, State disaster management authorities (SDMAs) oversee planning, capacity building and implementation of disaster mitigation measures while district disaster management authorities (DDMAs) focus on localized planning and operational response. This multi-tiered structure ensures that disaster management is decentralized and sensitive to regional hazards.

The policy emphasizes the importance of preparedness and capacity building. Early warning systems hazard mapping, vulnerability assessments and community based disaster preparedness programs are integral components of the framework. India has developed advanced meteorological and seismic monitoring systems to provide timely alerts for cyclones, floods and earthquakes helping to reduce casualities and property damage. Addition public awareness campaigns, training programs for emergency responders and mock drills are conducted regularly to enhance the preparedness of communities.

mitigation and risk reduction are central to India's disaster management approach. The policy promotes hazard-resistant infrastructure, land-use planning and sustainable development practices to reduce vulnerability. Special attention is given to protecting critical lifelines, including hospitals, schools and communication networks. Technological interventions such as Geographic information systems (GIS) and satellite imaging are increasingly employed for disaster risk mapping and

Planning. Furthermore the policy underscores the role of research, innovation and indigenous knowledge in developing effective mitigation strategies.
The financial and legal framework supporting disaster management is also robust. The Disaster management Act of 2005 provides the legal basis for establishing NDMA, SDMAs, DDMAs and mandates preparation of disaster management plans at all levels. The national disaster response fund (NDRF) and State disaster response funds (SDRFs) provide financial resources for immediate response and relief activities ensuring rapid deployment of assistance during emergencies.
Finally, India's disaster management policy emphasizes community participation and public-private partnerships. Empowering communities, involving non-governmental organization and fostering corporate social responsibility initiatives are crucial for building resilience. By integrating disaster management into developmental planning and promoting a cultural preparedness, the policy seeks to transform India into a disaster-resilient nation.
In Conclusion, the disaster management policy of India represents a comprehensive and integrated approach that combines prevention, mitigation, preparedness and response. Through institutional frameworks, technological tools, financial mechanisms and community engagement, the policy aims to minimize the impact of disasters, protect vulnerable populations and build long-term resilience. This forward looking strategy reflects India's commitment to ensuring safety, security and sustainable development in the face of diverse hazards.

Section- 'B'

Que 3. Write a note on manmade disasters.

Ans → manmade disasters are catastrophic events caused directly or indirectly by human actions, negligence or technological failures. unlike natural disasters, which are triggered by natural phenomenon, manmade disasters result from human activities and often have severe social, economic and environmental impacts. In India with its growing population, rapid industrialization and urban expansion, the risk of manmade disasters has increased significantly.

These disasters can be classified into several categories. Industrial accidents, such as chemical leaks explosions or fires in factories are common examples. The Bhopal Gas Tragedy of 1984 remains one of the most devastating industrial disasters in India, resulting in thousands of deaths and long term health consequences. Transportation accidents, including train derailments, road crashes and air disasters, also fall under this category, highlighting issues of infrastructure inadequacy and human error.

Another major category is environmental pollution caused by human activities. Industrial effluents, deforestation, improper waste management and oil spills degrade ecosystems. Sometimes resulting in disaster like toxic contamination or soil infertility. Similarly, nuclear accidents, such as the fukushima disaster also fall under this category highlighting issues of infrastructure inadequacy and human error.

Another major category is environmental pollution caused by human activities. Industrial effluents, deforestation improper waste management and oil spills degrade ecosystems sometimes resulting in disasters like toxic contamination or soil infertility. Similarly nuclear accidents, such as the Fukushima disaster in Japan, though not in India, Serve as a reminder of the Catastrophich potential of technological failures.

Terrorism and conflicts are increasing considered man made disasters due to their destructive effects on human life, property and Social Stability. Act of terrorism insurgency or communal riots disrupt communities and create long-term psychological and economic challanges. cyber disasters, though emerging also pose a significant threat, with attacks on critical infrastructure or financial systems causing widespread disruption.

The management of manmade disasters requires a proactive and comprehensive approach, India's disaster management Act of 2005 emphasizes preparedness, mitigation, response and recovery focusing on building resilient Communities and ensuring coordination among government agencies, NGOs and local populations. Risk assessment, strict enforcement of Industrial Safety standards, public awareness, campaigns and effective emergency response mechanisms are crucial to reducing the impact of such disasters.

In Conclusion, while manmade disasters are often preventable their consequences can be devastating if proper measures are not taken. Awareness, preparedness and responsible human actions are essential to minimize risk and safeguard lives, property and the environment.

Que. 4. What do you mean by vulnerability? Mention its determinants of identification.

Ans → Vulnerability refers to the degree to which individuals communities or systems are likely to experience harm due to exposure to hazards. It represents the susceptibility to damage, loss, or adverse effects when confronted with natural or man-made disasters. Vulnerability is not solely determined by the Hazard itself but is a function of social economic, environmental and institutional conditions that affect the capacity of people to anticipate, cope with resist and recover from the impact of a disaster. In simple terms vulnerability reflects the weakness or fragility that increases the risk of disaster losses.

Identifying vulnerability is a crucial step in disaster management because it helps in prioritizing interventions, resource allocation and planning for risk reduction. Several determinants are considered for its identification:

1. Physical Determinants: These relate to the characteristics of the environment and infrastructure that make people or assets susceptible to harm. Examples include living in flood-prone areas, poor housing construction, proximity to earthquakes zones, or lack of protective barriers.

2. Economic Determinants: Economic vulnerability arises from poverty, lack of income and independence on limited livelihood options. Poor communities often lack savings, insurance or access to financial resources which restricts their capacity to recover after a disaster.

3. Social Determinants: Social factors include age, gender education, health and social status. Children, elderly people and persons with disabilities are more vulnerable due to limited mobility and dependence on others. Low literacy and lack of awareness about disaster preparedne further increase vulnerability.

4. Institutional Determinants: These include the effectiveness of governance, public services and disaster management institutions. Weak administrative structures, inadequate enforcement of building codes and poor emergency service amplify vulnerability.

5. Environmental Determinants: Degradation of natural resources, deforestation, soil erosion and climate change related factors contribute to environmental vulnerability. These factors can intensify the impact of hazards such as floods, landslides and droughts.

6. Technological Determinants: Vulnerability can also arise due to dependency on technology industrial hazards, or unsafe infrastructure. Accidents in chemical plants or nuclear facilities are examples where technological determinants play a role.

Understanding these determinants enables disaster managers and policymakers to design targeted interventions to reduce vulnerability. Strengthen resilience and promote sustainable development. Effective disaster management requires integrating these insights into planning, preparedness and mitigation Stertegies.

Que. 5. Discuss various types of natural disasters that occur in India.

Ans → India, due to its unique geographical location and diverse climatic conditions, is highly prone to a variety of natural disasters. These disasters affect millions of people each year causing loss of life, property damage and disruption of livelihoods. understanding the different types of natural disasters in India is crucial for effective disaster management.

1. Earthquakes :- India lies in a seismically active zone especially in the Himalayan region, parts of Northeast India and the Indo-Gangetic plains. The collision of the Indian plate with the Eurasian plate makes the northern region particularly vulnerable. Earthquakes can lead to building collapses, landslides and infrastructure damage, affecting urban and rural populations alike.

2. Floods :- Flooding is one of the most frequent disasters in India, primarily caused by heavy monsoon rains, river overflow, and cyclonic storms. States like - Assam Bihar, uttar pradesh and west Bengal are highly flood-prone. Floods result in displacement, loss of crops water contamination and outbreaks of water-borne diseases. urban areas also face flash floods due to poor drainage systems.

3. Cyclones :- The eastern and western coasts of India particularly odisha, Andhra pradesh, Tamil Nadu and west Bengal are vulnerable to cyclones originating in

the Bay of Bengal and the Arabian Sea. Cyclones bring strong winds, heavy rainfall and storm surges leading to widespread destruction of property, crops and sometimes human lives.

4. Droughts :- Droughts are common in the western and central parts of India, such as Rajasthan, Gujarat and parts of maharashtra. They result from prolonged periods of deficient rainfall, leading to water scarcity, crop failure and depletion of groundwater Droughts severly affect agriculture, livestuck and rural livelihoods.

5. LandSlides :- Hilly regions such as the Himalayas, western Ghats and the northeastern States experience landslides, particulary during the monsoon. Heavy rainfa deforestation and soil erosion trigger landslides, which can bury villages, roads and infrastructure, causing loss of life and economic setbacks.

6. other Disasters :- India also faces other natural hazards such as heatwaves, cold waves, avalanches and tsunamis The 2004 Indian ocean tsunami highlighted the vulnerability of coastal regions to such rare but catastrophic events.

In conclusion, India's vulnerability to multiple natural disasters necessitates a robust disaster management system involving preparedness, mitigation response and recovery measures to minimize the impact on human lives and economic stability.

Section - 'C'

Que. 6. Define the term Hazard.

Ans → A hazard can be defined as a naturally occurring or human induced phenomenon that has the potential to cause harm to people, property or the environment. It represents a source of danger that can result in physical damage, economic loss, or social disruption. Hazards can be classified into various types such as natural hazardous – including earthquakes, floods, cyclones and landslides – and human made hazards like industrial accidents, nuclear disasters and chemical spills. The intensity and frequency of a hazard determine its potential threat level. Hazards are not disasters by themselves; they become disasters only when they interact with vulnerable populations, inadequate infrastructure or insufficient preparedness. understanding the nature, origin and characteristics of hazards is essential for disaster management, as it enables authorities and Communities to implement preventive and preparedness measures. Risk assessment often involves identifying hazards, estimating their probability and evaluating the potential impact on Society. Effective hazard management reduces vulnerability and enhances resilience, ensuring the hazards do not escalate into major disasters.

Que 7. What do you mean by hydrological Disasters?

Ans → Hydrological disasters are natural events that result from the occurence, movement and distribution of water in the environment, leading to significant damage to life property and ecosystems. These disasters are directly related to water-related processes, such as rainfall, river flow, and oceanic or groundwater systems. Common examples include floods, flash floods, storm surges, tsunamis and droughts. Floods occur when water bodies exceed their capacity, inundating surrounding areas, while droughts represent prolonged periods of deficient rainfall, affecting agriculture, water supply and livelihoods. Hydrological disasters often disrupt infrastructure, damage crops and increase the risk of water borne diseases. Climate change, urbanization, deforestation and poor water management practices have intensified the frequency and severity of such events. Disaster management strategies for hydrological hazard include early warning systems. Construction of embankments and reservoirs, watershed management and public awareness programs, understanding the hydrological cycle and regional water patterns is crucial to predict, mitigate and manage these disasters effectively.

Que 8. Mention disaster mitigation approaches.

Ans → Disaster mitigation refers to measures taken to reduce the impact of natural or human-induced disasters before they occur. mitigation approaches aim to minimize human, economic and environmental losses. These approaches can be structural or non-structural. Structural mitigation includes physical measures like constructing dams, embankments, earthquake resistant buildings cyclone shelters and improved drainage systems. Non-structural mitigation involves policies, regulations and community based strategies, such as land-use planning hazard mapping, awareness compaigns, early warning systems and training for emergency response. Environmental management like afforestation and soil conservation is another mitigation approach that reduces vulnerability to floods and landslides. Social and institutional measures such as establishing disaster management authorities and emergency funds, also strengthen mitigation capacity. Risk assessment, vulnerability analysis, and scenario planning form the backbone of mitigation strategies. A combination of these approaches ensures a holistic reduction in disaster risks, building resilient communities capable of withstanding natural and human-made hazards.

Que. 9. Mention various dimensions of damage assessment.

Ans → Damage Assessment is a systematic evolution of the losses caused by a disaster and provides the foundation for recovery planning and resource allocation. various dimensions of damage assessment include physical, economic Social and environmental aspects. Physical assessment focus on the deconstruction of infrastructure, buildings, roads, bridges and communication networks. Economic assessments evaluates directs and indirect financial losses such as damage to crops, livestock, industries and livelihoods. Social assessments examines the impact on human lives, health, education and displacement of communities, addressing issues like mortality, injuries and loss of Shelter. Environmental assessment Considers the degradation of natural resources, Soil erosion, deforestation water Conteminartion and ecological imbalance caused by the disaster. Effective damage assessment also includes spatial and temporal analysis to understand the geographical extent and duration of impacts. Accurate damage assessment is essential for planning rehabiliation mobilizing resources, insurance, claims and implementing disaster risk reduction measures to prevent future losses.

Que 10. Define rehabilition and bring out its typology.

Ans → Rehabiliation is the process of restoring normalcy and improving the living conditions of individuals and communities affected by a disaster. It involves reconstructing infrastructure restoring livelihoods and providing Psychological and social Support to ensure that affected populations can resume their daily activities. Rehabilition is distinct from immediate relief; it focuses on long-term recovery and building resilience against future disasters. The typology of rehabilition includes physical, economic, Social and Psychological dimension Physical rehabiliation deals with the reconstruction of houses school, roads, and public utilities. Economic rehabiliation involves restoring livelihoods, employment opportunities and financial support to affected communities. Social rehabiliation emphasizes restoring social networks. community cohesion and access to essential services such as healthcare and education. Psychological rehabilition addresses mental health trauma counseling, and emotional well-being. Comprehensive rehabilition integrates all these dimensions to ensure that communities not only recover from disasters but also become more resilient and prepared for future hazards.